NAME: NEEMA MAGABE

REG NO:22/00135

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SUPERVISOR: RISPER KENDI

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INTRODUCTION

1.1 Purpose.

The purpose of the system is to analyze customer data and predict whether a customer is likely to churn (leave the telecom service). The system will assist the company in identifying high-risk customers and improving retention strategies.

1.2 Intended Audience and Reading Suggestions

The document is intended for the following stakeholders

Project Supervisor/Lecturer**:** project supervisor will evaluate the system requirements, design clarity, and technical completeness of the churn prediction system.

Developers/Data Scientists: they will be required to understand the functional and non-functional requirements necessary for implementing the system.

Telecom Business Analysts: they will understand how the system supports customer retention strategies through churn prediction.

System Testers – To verify that the system meets specified requirements and performance criteria.

1.3 Scope

The Telecom Customer Churn Prediction System is a machine learning-based software application designed to analyze historical customer data and predict the customer churn. The system uses features such as customer tenure, service subscriptions, contract type, monthly charges, and payment methods to generate predictive insights.

The primary purpose of this system is to assist telecom companies in identifying customers who are at high risk of leaving the service. By predicting churn in advance, the organization can implement targeted retention strategies such as personalized offers, improved customer support, and contract adjustments.

The key objectives of the system include:

- Developing accurate predictive models (e.g., Logistic Regression and Random Forest) to classify churn and non-churn customers.
- Providing performance evaluation metrics such as Accuracy, Precision, Recall, and F1-score.
- Supporting data-driven decision-making for customer retention.
- Reducing revenue loss caused by customer attrition.

The system aligns with corporate business strategies focused on customer retention, revenue optimization, and competitive advantage. Since acquiring new customers is generally more expensive than retaining existing ones, the implementation of a churn prediction system supports long-term sustainability and operational efficiency within the telecom industry.

2. Overall Description

2.1 Product Perspective

The Telecom Customer Churn Prediction System is a new software application developed to help telecom companies predict which customers are likely to leave their services. It is a standalone system built for analysis purposes and does not replace any existing system.

Currently, telecom companies use systems like CRM and billing systems to store customer information, manage payments, and handle services. However, these systems do not predict future customer behavior. This project adds value by analyzing stored customer data and predicting churn using machine learning models.

The system works by:

- Taking customer data from Excel or CSV files
- Processing and cleaning the data
- Training machine learning models
- Predicting whether a customer is likely to churn
- Displaying results such as accuracy, precision, recall, F1-score, and confusion matrix

In a real organization, this system can support business strategies such as improving customer retention and reducing revenue loss.

3 system requirements

3.1 Product Functions (functional requirement)

The Telecom Customer Churn Prediction System provides several important features that help analyze customer data and predict churn, like:

1. Data Upload Feature

The system allows the user to upload customer data in Excel or CSV format. This data contains customer details such as tenure, contract type, services subscribed, and payment information.

2. Data Preprocessing Feature

The system cleans and prepares the data for analysis. This includes handling missing values, converting categorical data into numerical format, and organizing the dataset for training.

3. Model Training Feature

The system trains machine learning models such as:

- Logistic Regression
- Random Forest

4. Model Evaluation Feature

The system evaluates model performance using:

- Accuracy
- Precision
- Recall
- F1-score

This helps determine how well the model performs.

6. Prediction Feature

The system allows the user to input new customer data and predicts whether the customer is likely to churn.

3.2 Non-functional requirement

- Data Privacy

  Protect customer personally identifiable information (PII) such as phone numbers, addresses, or account numbers.
- Data Integrity

  Prevent loss or corruption of customer data during preprocessing or analysis.
- Non-destructive Processing

  Ensure that original datasets are never overwritten or lost.
- Error Handling

  Avoid incorrect model predictions or misleading results due to bad data or system errors.

- Backup and Recovery

  Prevent permanent loss of datasets or models due to accidental deletion or system failure.

3.3 User Classes and Characteristics

The Telecom Customer Churn Prediction System will be used by different types of users. Each user has different responsibilities, technical skills, and access levels.

1. Administrator

The Administrator is usually an IT specialist or system manager responsible for managing and maintaining the system.

Characteristics:

- Has strong technical knowledge (IT or data science background).
- Uses the system frequently.
- Has full access and control over system functions.
- Responsible for uploading datasets, running models, and managing system settings.

Responsibilities:

- Upload and manage customer data.
- Train and evaluate models.
- Monitor system performance.
- Ensure data security and integrity.

2. Data Analyst

This user focuses on analyzing results and making business decisions based on predictions.

Characteristics:

- Has moderate to strong knowledge of data analysis.
- Uses the system regularly.
- Accesses reports and performance metrics.
- May not manage system configurations.

Responsibilities:

- Interpret model results.
- Identify high-risk customers.
- Support customer retention strategies.
- Generate reports for management.

Clerks

These users may use the system occasionally to check churn predictions for specific customers.

Characteristics:

- Limited technical knowledge.
- Use only basic features.
- Limited access privileges.

Responsibilities:

- View churn predictions.
- Input new customer details for prediction.
- Support customer service operations.

3.4 Operating Environment

The Telecom Customer Churn Prediction System will operate in a standard computing environment suitable for data analysis and machine learning tasks.

Hardware Environment

- A personal computer or laptop.
- Minimum 4GB RAM
- Standard keyboard, mouse, and monitor.

Software Environment

- Operating System: Windows 11
- Programming Language: Python.
- Development Environment: Jupyter Notebook, VS Code, or any Python IDE.
- Required Libraries:
    - pandas (for data manipulation)
    - NumPy (for numerical computation)
    - scikit-learn (for machine learning models)
    - matplotlib / seaborn (for visualization)

Data Environment

- Input data will be provided in Excel (.xlsx) or CSV format.
- The dataset will contain historical telecom customer information used for training and testing the models.

3.5 Design and Implementation Constraints

The Telecom Customer Churn Prediction System is subject to certain design and implementation constraints that may affect how the system is developed and deployed.

1. Technology Constraints

- The system must be developed using Python**.**
- Machine learning models must be implemented using libraries such as scikit-learn**.**
- The system must accept input data in Excel (.xlsx) or CSV format.
- Development will be carried out using tools such as Jupyter Notebook or VS Code.

2. Hardware Constraints

- The system will run on a standard personal computer.

3. Data Constraints

- The system depends on the quality and completeness of historical customer data.
- Missing, incorrect, or imbalanced data may affect prediction accuracy.
- The dataset structure must remain consistent

4. Security Constraints

- Customer data must be handled confidentially.
- Access to the system should be restricted based on user roles

4. Appendix

- **Churn:** The event of a customer leaving the telecom service provider.
- **CustomerID** Unique identifier assigned to each customer.
- **EDA (Exploratory Data Analysis):** The process of analyzing datasets to summarize their main characteristics, often with visual methods.
- **CSV (Comma-Separated Values):** A simple file format used to store tabular data, where values are separated by commas.
- **Python:** Programming language used for data preprocessing, analysis, and model building.
- **Scikit-learn:** Python library for machine learning.
- **Matplotlib / Seaborn:** Python libraries used for data visualization.
- **NumPy:** python library for mathematical numeric.

- **Pandas:** python library for uploading files

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